PARISHA JOSHI

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SUMMARY

I am developing accurate and computationally efficient models to reconstruct spectral properties of illuminants. **My research enhances the data driven inverse rendering pipelines for accurate structural color renderings such as diffraction, refraction, and iridescence**. Which is to be used in simulated environments, computer vision, and 3D space reconstructions.

EDUCATION

Clemson University Clemson, SC
Ph.D Computer Science Expected Dec 2026

- *Coursework:* Advanced Data Structures, Advanced Machine Learning, Data Mining, Cloud Computing
- Advised by Dr. Daljit Singh Dhillon. Published in SIGGRAPH, ICVGIP, Pacific Graphics, Marketing director-SOC Grad Student Association,Spring 2025.

Virginia Tech Blacksburg, VA
Master's Computer Engineering Aug 2020

- *Coursework:* Computer Vision, Experimental Robotics , Network Arch and Protocols
- Best Summer Research Award: UCAL-JAP, Built wildfire spread prediction using hyperspectral images framework working with Dr.Brian Lattimer.

Pandit Deendayal Energy University Gandhinagar, India
B.Tech Electrical Engineering May 2017

- *Coursework:* Optimization techniques, Signal Processing, Numerical and Statistical Methods
- Motor Control department-Shell Eco Marathon with Team Kaizen, Treasurer-IEEE Student Chapter, Technical Chair-Mind Ripple Quiz Club

WORK EXPERIENCE

Graduate Research Assistant Aug 2022 – Present
Clemson University *Clemson, SC*

- Worked on inverse rendering of structural materials predicting illumination SPD in the 2D to 3D scene reconstruction.
- Worked on robust color correction of images using color-chart spectral densities using gradient descent algorithm.
- Contributed to VarIS: Variable Illumination Sphere, creating calibrated datasets for material capture using ChAruco Board.
- Data visualization and analysis project on reaction-diffusion systems using gray-scott equation.
- Implemented anisotropic BRDF models, including the Kajiya–Kay and Marschner frameworks, to render realistic hair materials.
- Adapted the method proposed in Matsoukas at el. to improve segmentation accuracy on traffic image datasets.

Full-Stack Java Developer Aug 2020 – Jul 2021
MPower Logic, Inc – StateFarm *Bloomington, IL*

- Designed front-end services with ReactJS and REST APIs for dynamic, data-sensitive customer interactions.
- Applied A/B testing and feedback analysis to optimize apps, skills transferable to ML system tuning.
- Built scalable components with cloud integration, enabling rapid CI/CD in an Agile development workflow.

Graduate Research Assistant Aug 2019 – Aug 2020
Extreme Environments and Materials Lab, Virginia Tech *Blacksburg, VA*

- Built a ROS-based sensor fusion system with LiDAR and RGB-D camera for a ground robot to map lab space in real-time and deliver coffee to QR-tagged tables. Used ROS, PCL, OpenCV, Python, and Gazebo.
- Designed a wildfire prediction system using Transpose CNNs, achieving 96% accuracy on spatiotemporal data.
- Managed multi-spectral and environmental data to train and validate real-world forecasting models.
- Optimized model performance using Adam optimizer and dropout regularization, reducing validation loss by 18%.

Research Intern May 2019 – Aug 2019
UCAL-JAP Systems Ltd *Chennai, India*

- Engineered a drone simulation for object detection using Python, ROS, and ML for autonomous systems.
- Implemented a YOLOv3-based real-time object detection model, achieving over 85% MAP on drone imagery.

Embedded Systems Engineer August 2017 – December 2017
National Infotech *Surat, India*

- Designed and programmed an electric BLDC dynamic motor controller using the STM32F4 microcontroller.
- Analyzed hardware architecture and motor performance under loaded and unloaded conditions through simulations.

Electrical Support Engineer Intern June 2016 - July 2016
Transformers and Rectifiers *Ahmedabad, India*

- Project: *Strategic Use of IT for SLA Monitoring and Control Center System*.

- Gained hands-on exposure to the manufacturing process of power, distribution, and furnace transformers.
- Developed self-reliance, initiative, and effective communication skills through interactions with engineers, workers, and supervisors.

Electrical Support Engineer Intern June 2016 – July 2016
Transformers and Rectifiers *Ahmedabad, India*

- Project: *Strategic Use of IT for SLA Monitoring and Control Center System*.
- Gained hands-on exposure to the manufacturing processes of power, distribution, and furnace transformers.
- Strengthened self-reliance, initiative, and communication skills through close collaboration with engineers, workers, and supervisors.

PROJECTS

Extracting Information from Reaction-Diffusion Patterns (Project Link) Jan 2023 – May 2023
Extracting statistical information from Gray-Scott model patterns using ParaView.

- Used Python to implement the Gray-Scott model which forms biological patterns such as spots and stripes on animals, flowers and fishes.

Realistic hair in PBRT (Project Link) Aug 2021 – Dec 2021
3D Rendering of realist hair using PBRT with Marschner's and Kajiy-Kay's model

- Used C++ to extend the pbrt library with two additional materials named ”kajiya” and ”marschner”.

Thin-Film Interference (Project Link) Aug 2021 – Dec 2021
Using PBRT and C++, modeled Thin-film BRDF.

- Rendering bubbles with air-soap-air layers.

Segmentation task of traffic images in low data regime (Project Link) Aug 2021 – Dec 2021
Implementation with Deep Convolutional Inverse Graphics Neural Network(DCIGN)

- Explored convolutional autoencoders and inverse graphics neural networks to handle limited, unstructured data for traffic segmentation tasks.
- Compared model outputs using accuracy, precision, recall and F1 Score, refining hyperparameters for robust detection under sparse training conditions.

Credit Card Fraud Detection (Project Link) May 2023 – July 2023
Fraud detection, XGBoost/Random Forest, ROC-AUC

- Developed fraud detection models using Random Forest, XGBoost, and Logistic Regression on labeled financial data.
- Integrated risk analytics and model validation to quantify financial impact and enhance fraud detection reliability.

Predicting Hearing Loss Threshold from Brain MRI (Project Link) Aug 2023 – Dec 2023
Regression modeling, CNN/MLP, Healthcare analytics

- Built CNN model to predict hearing thresholds from MRI scans with 89% accuracy using limited sample sizes.
- Designed reproducible pipelines for feature extraction and evaluation.
- Trained on Python-Keras library, applications in biometric scoring and clinical AI tools.

Time Series Forecasting for Stock Prices (Project Link) May 2021 – July 2021
Forecasting, LSTM/GRU, Risk modeling

- Implemented LSTM and GRU-based forecasting models for financial time series forecasting.
- Incorporated market sentiment and macroeconomic signals as additional features.
- Calibrated and validated models using historical data to improve predictive accuracy in financial risk modeling.

Fake News Detection Using ML (Project Link) May 2022 – July 2022
NLP, TF-IDF, Random Forest

- Built a Random Forest model to classify news articles as fake or real using TF-IDF and textual features.
- Designed an end-to-end pipeline for text preprocessing,
- Used feature engineering, and classification to boost precision in misinformation detection by 12%.

Reinforcement Learning: Policy Iteration (Project Link) Aug 2019 – Dec 2019
Policy optimization, Simulation modeling, MATLAB

- Simulated policy iteration under uncertainty to evaluate reward-cost tradeoffs in decision-making scenarios.
- Applied Markov Decision Process (MDP) to model optimal actions in constrained environments.

Direction Of Arrival Estimation (Project Link) Aug 2019 – Dec 2019
Use of Deep Neural Network(DNN)

- Built a deep neural network model for estimating the direction of arrival(DOA) of 2,3 and 4 signals. Use of MATLAB GUI tools for signal processing.
- Compared with MUSIC estimator and checked with CRLB values for both in Noiseless and SNR range from 0 to 10 dbs.

SKILLS

Programming	Python, R, Java, C++, SQL, JavaScript, ReactJS
AI/ML	GNN, Reinforcement Learning, Random Forests, XGBoost, LSTM, MLP, CNN, NLP , RNN
Software/Tools	ML platforms Tensorflow/Keras, PyTorch, Jupyter, Google Colab, Linux, BASH scripting.
Version Control	Git, GitHub/GitLab, BitBucket, Bamboo, Jira

TEACHING EXPERIENCE

Graduate Teaching Assistant Aug 2021 – Present
Clemson University *Clemson, SC*

- CPSC 8420 : Advanced Machine Learning by Dr. Kai Liu, Fall 2025 (TA)
- CPSC 3120 : Design of Algorithms by Dr. Nicolas Widman, Fall 2024 (TA)
- CPSC 2070 : Discrete Structures by Dr. Nicolas Widman, Spring 2022 (TA)
- CPSC 1210 : Computational thinking by Prof. Mitch Shue, Fall 2021 (TA)

Graduate Teaching Assistant Aug 2018 – Aug 2020
Virginia Tech *Blacksburg, VA*

- ECE 5560 : Fundamentals of information security by Dr. Jerry Park, Fall 2019 (TA)
- ECE 2004 : Electric Circuit Analysis by Dr. Mona Ghassemi, Fall 2018 (Grader)
- Math 1014 : Pre-calculus with Transcendental Functions, Fall 2018 (Tutor)
- Math 1535 : Geometry and the Mathematics of Design, Fall 2018 (Tutor)

Tutoring Aug 2017 – Aug 2018
Disha - The footpath School *Surat, Gujarat, India*

- Worked with Disha - The footpath School for one year teaching Maths and Science to under privileged students.

Volunteering Aug 2016 – Aug 2017
Swami Rajchandra Adhyatmik Sadhna Kendra *Koba, Gujarat, India*

- Took the initiative to teach mathematics to underprivileged kids from grades 5th to 7th.

PUBLICATIONS & CERTIFICATION

- **Joshi Parisha,** Dhillon Daljit Singh J , *"Accurate Spotlight Spectra Estimation Using Neural Methods and Compact Disc Diffraction Gratings"* In Proceedings

- **Joshi Parisha,** Dhillon Daljit Singh J , *"Learning-Based Accurate Estimation of Spot-light Illuminant Spectra using Diffraction Gratings on CD-ROM"* The Eurographics Association (Spanish Computer Graphics Conference (CEIG)) Valencia Spain

- Dhillon Daljit Singh J, **Joshi Parisha** , *"Improving Convergence of Fourier Optical Shading for Nanostructural Diffraction"* SIGGRAPH 2026 Posters Los Angeles USA

- **Joshi Parisha**, Dhillon Daljit Singh , *"Practical and Accurate Reconstruction of an Illuminant's Spectral Power Distribution for Inverse Rendering Pipelines"* ICVGIP2024 Bangalore India `https://arxiv.org/abs/2410.22679`

- Daljit Singh Dhillon, **Joshi Parisha**, Jessica Baron and Eric K Patterson., *"Robust color correction for preserving spatial variations in photographs", Poster* Siggraph'23 Los Angeles USA `https://doi.org/10.1145/3588028.3603681`

- Daljit Singh Dhillon, **Joshi Parisha**, Jessica Baron and Eric K Patterson., *"Robust Distribution-aware Color Correction for Single-shot Images"* Pacific Graphics'23 Daejeon South Korea `https://doi.org/10.1111/cgf.14959`

- Baron Jessica, Li Xiang, **Joshi Parisha**, Itty Nathaniel, Greene Sarah, Dhillon Daljit Singh J. and Patterson Eric, *"VarIS: Variable Illumination Sphere for Facial Capture, Model Scanning, and Spatially Varying Appearance Acquisition"*
STAG2023, The Eurographics Association Matera Italy `https://doi.org/10.2312/stag.20231292`

- Parisha Maheshbhai Joshi, ***"AWS Certified Cloud Practitioner"*** Validation Number c0c8fe681f6f42198fec2fba90d71987 Active Date : 2025-05-31 Expire Date : 2028-05-31 Validate on 'https://cp.certmetrics.com/amazon/en/public/verify/credential'

- Parisha Maheshbhai Joshi, ***" NVIDIA OpenACC – 2X in 4 Steps",*** August 2020

- Parisha Maheshbhai Joshi, ***"Tableau for Data Science",*** May 2020